

Note 15.6 Moments and Centers of Mass

Masses and First Moments

Mass

The mass of a object with $\delta(x, y, z)$ as the density is

$$M = \iiint_D \delta(x, y, z) dV$$

First Moment

The first moment is defined as the multiplication of the mass of the object and the distance from the object to the rotation plane(rotation axis in two dimensions), or, the triple integral(double integral in two dimensions) over D of the distance from the point (x, y, z) to the plane multiplied by the density of the solid at the point.

Center of Mass

The center of mass is found from the first moments, defined by first moment over total mass.

For instance, the x -coordinate of the center of mass is $\bar{x} = M_{yz}/M$.

Centroid

When the density of a solid object or plate is constant, the center of mass is called the centroid of the object.(same to the geometric center)

Moments of Inertia

The moment of inertia(or the second moment) focus on how much energy is stored in the shaft or how much energy is generated by rotating at a particular angular velocity.

The moment of inertia of the shaft about its axis of rotation is

$$I = \int r^2 dm = \int r^2 \delta dV$$

And the kinetic energy is

$$\text{KE}_{\text{shaft}} = \frac{1}{2} I w^2$$

where r is the distance to the axis of rotation, w is angular velocity.

The moments of inertia about the line L in three dimension is:

$$I_L = \iiint r^2 \delta dV$$

In two dimension is

$$I_L = \iint r^2 \delta dV$$

Note that in two-dimensional plate, there is a moment of inertia about the origin that is

$$I_0 = I_x + I_y = \iint (x^2 + y^2) \delta dA$$